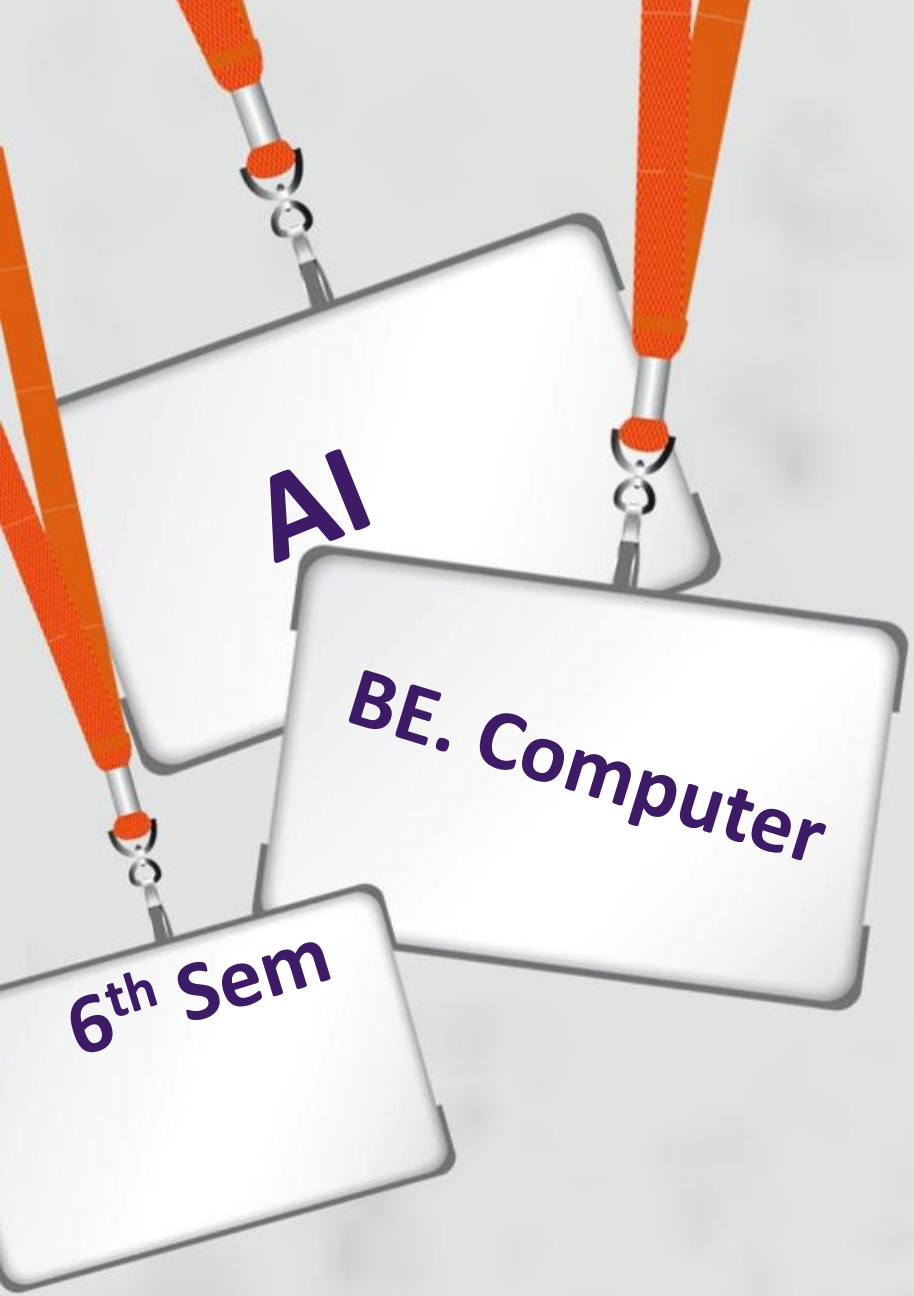


Chapter 5

Structured Knowledge Representation

Knowledge

- Knowledge is a theoretical or practical understanding of a subject or a domain and it is also the sum of what is currently known.
- Hence, knowledge is the sum of what is known: the body of truth, information, and principles acquired by mankind.
- Knowledge according to Sunasee and Sewery, 2002,

"*Knowledge is human proficiency stored in a person's mind, gained through experience, and interaction with the person's environment."*

Knowledge

- In general, knowledge is more than just data, it consist of: facts, ideas, beliefs, heuristics, associations, rules, abstractions, relationships, customs.
- Different types of Knowledge:
  - Declarative Knowledge
  - Procedural Knowledge
  - Heuristic knowledge
  - Meta-knowledge
  - Structural knowledge

Knowledge Types (1)

- **Declarative Knowledge:**
  - Declarative knowledge is to know about something.
  - It includes concepts, facts, and objects.
  - It is also called descriptive knowledge and expressed in declarative sentences.
  - It is simpler than procedural language.

- **Procedural Knowledge:**
  - It is also known as imperative knowledge.
  - Procedural knowledge is a type of knowledge which is responsible for knowing how to do something.
  - It can be directly applied to any task.
  - It includes rules, strategies, procedures, agendas, etc.
  - Procedural knowledge depends on the task on which it can be applied.

Knowledge Types (2)

- **Heuristic Knowledge:**
  - Heuristic knowledge is representing knowledge of some experts in a filed or subject.
  - Heuristic knowledge is rules of thumb based on previous experiences, awareness of approaches, and which are good to work but not guaranteed.
  - In this type, the knowledge representation is based on the strategies to solve the problems through the experience of past problems, compiled by an expert. Hence, it is also known as **Shallow knowledge**
- **Meta Knowledge:**
  - Knowledge about the other types of knowledge is called Meta-knowledge.

Knowledge Types (3)

- **Structured Knowledge**:
  - Structural knowledge is basic knowledge to problem-solving.
  - It describes relationships between various concepts such as kind of, part of, and grouping of something.
  - It describes the relationship that exists between concepts or objects.

Knowledge Representation

- Humans are best at understanding, reasoning, and interpreting knowledge.
- Human knows things, which is knowledge and as per their knowledge they perform various actions in the real world.
- **But how machines do all these things comes under knowledge representation and reasoning.**

Knowledge Representation

- Knowledge representation (KR) is the study of how knowledge about the world can be represented and what kinds of reasoning can be done with that knowledge.
- Basically, it is a study of how the **beliefs, intentions**, and **judgments** of an **intelligent agent** can be expressed suitably for automated reasoning.
- Knowledge Representation is the method used to encode knowledge in Intelligent Systems.

Knowledge Representation

- It is responsible for representing information about the real world so that a computer can understand and can utilize this knowledge to solve the complex real world problems such as diagnosis a medical condition or communicating with humans in natural language.
- Knowledge representation is not just storing data into some database, but it also enables an intelligent machine to learn from that knowledge and experiences so that it can behave intelligently like a human.

Issues in Knowledge Representation

Some issues that arise in knowledge representation from an AI perspective are:

- How do people represent knowledge?
- What is the nature of knowledge and how do we represent it?
- Should a representation scheme deal with a particular domain or should it be general purpose?
- How expressive is a representation scheme or formal language?
- Should the scheme be declarative or procedural?

Properties of KR System

- **Representational Adequacy:**
  - It is the ability of the system to represent all kinds of knowledge needed in a specific domain.
- **Inferential Adequacy:**
  - It is the ability of a knowledge representation system to manipulate the current stored knowledge so that newly gained knowledge could be added.
- **Inferential Efficiency:**
  - It is the ability of the system to directly add new knowledge in the system with efficiency
- **Acquisitional Efficiency:**
  - It is the ability of the system to automatically acquire new knowledge from the environment.
  - This leads the system to give more productive result as more knowledge adds up with the current knowledge.

Knowledge Representation Techniques

- Different types of KR techniques are:
  - Semantic Network
  - Frames
  - Conceptual Dependencies
  - Scripts

Semantic Network (1)

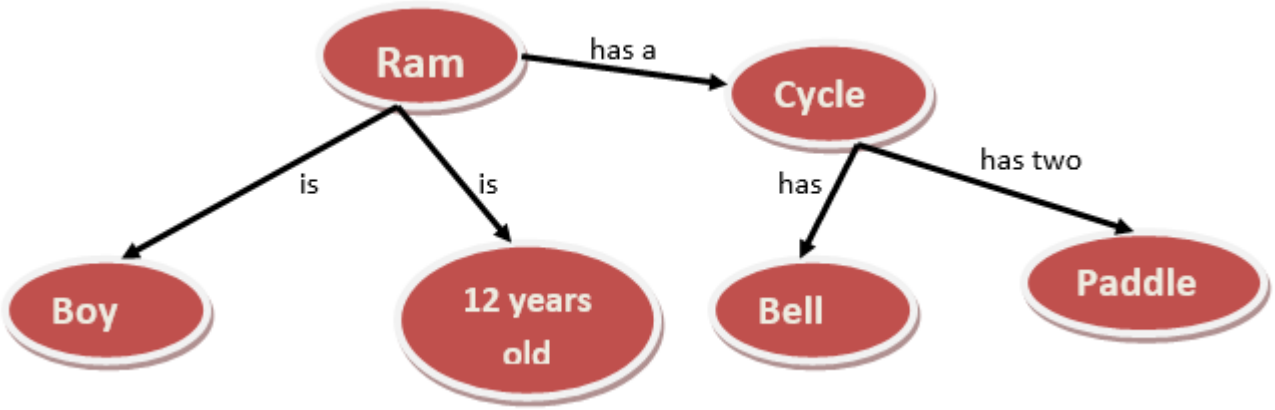
- It is classic AI representation technique used for propositional information.
- Also Known as proposition nets.
- The technique is based on storing the knowledge into the system in the form of a graph.
- Nodes of a graph represent the objects which exist in the real world, and the arrow represents the relationship between these objects.
- Such techniques show the connectivity of one object with another object

Semantic Network (2)

- A semantic net is a labelled, directed graph The structure of a semantic net is shown graphically in terms of nodes and the arcs connecting them.
  - Nodes are sometimes referred to as objects
  - Arcs as links or edges are used to express relationships.
- A node can represent a fact description
  - Physical object
  - Concept
  - Event
- An arc (or link) represents relationships between nodes. There are some 'standard' relationship types
  - 'Is-a' relationship
  - 'Has-a' relationship

Semantic Network (3)

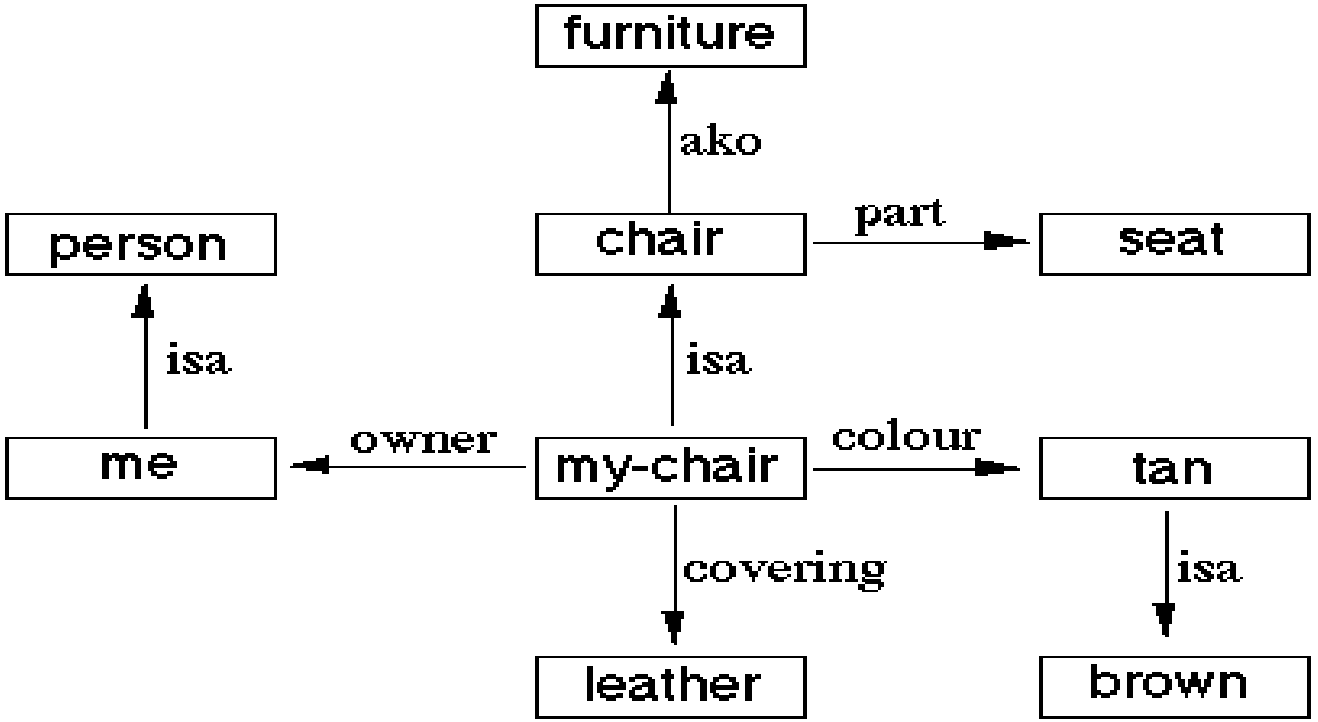
- Consider the given knowledge:
  - Ram has a cycle.
  - Ram is a boy.
  - Cycle has a bell.
  - Ram is 12 years old.
  - Cycle has two paddles.



Simple Semantic Network

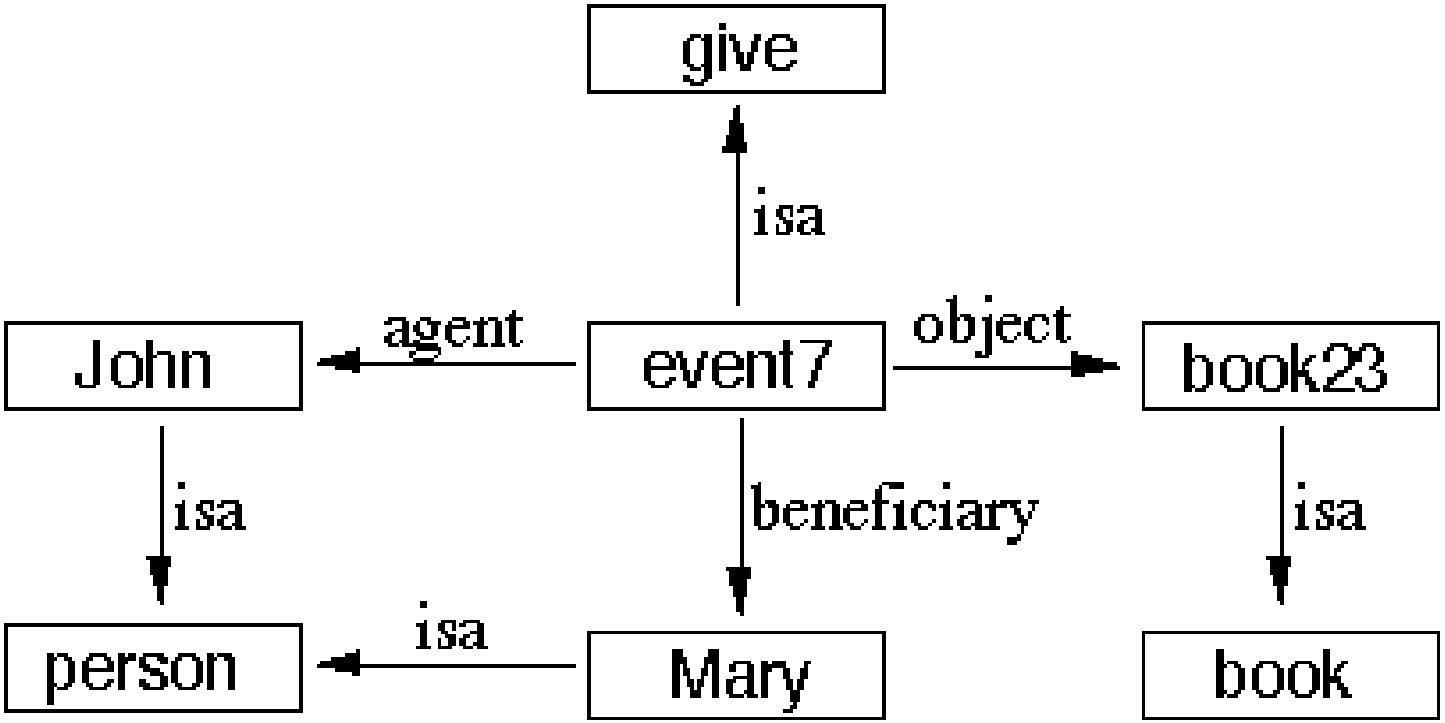
Semantic Network (4)

I own a tan leather chair



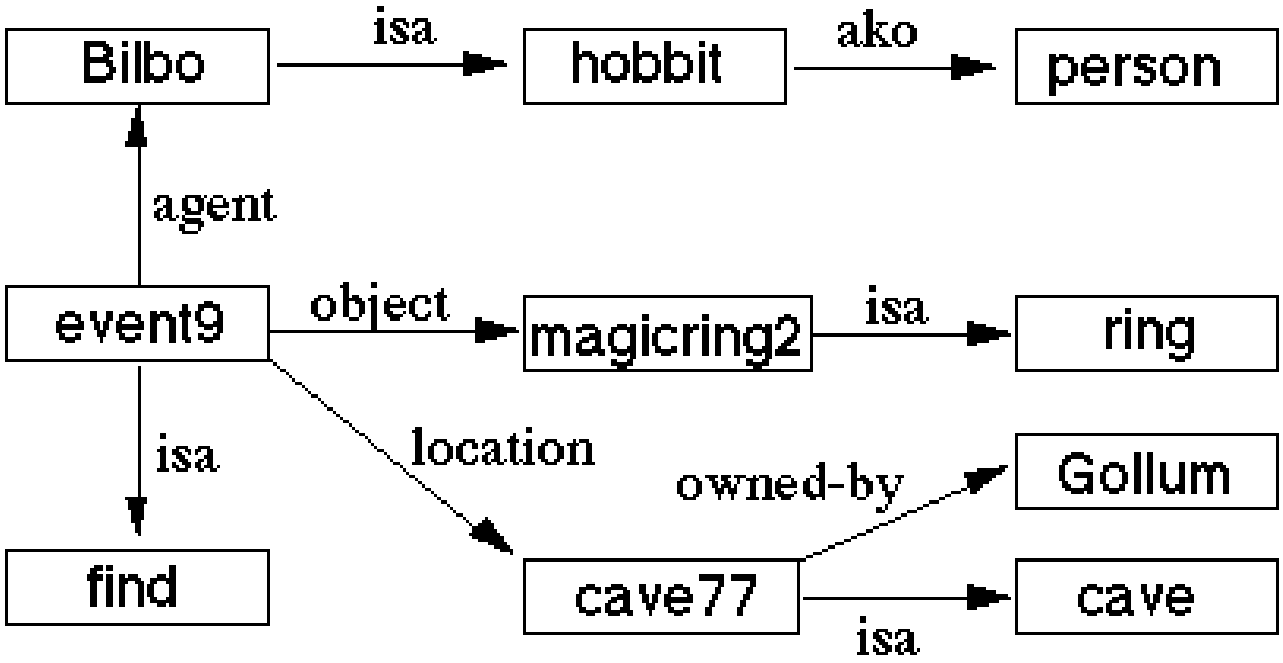
Semantic Network (5)

John gives the book to Mary



Semantic Network (6)

Bilbo finds the magic ring in Gollum's cave



Advantages of Semantic Networks

- They tend to be a powerful and adaptable method of representing knowledge because many different types of object can be included in the network.
- The network is graphical and therefore relatively easy to understand.
- Can be used as a common communication tool between the knowledge engineer and the human expert during the knowledge acquisition phase of designing an ES.

Disadvantages of Semantic Networks

- It can be difficult to show all the different inference situations using a network.
- They are less reliable than other knowledge representation techniques because inferring becomes a process of searching across the diagram.
- Diagrams can become very complex i.e. the wide range of possible kinds of links and the ways they might combine to form indirect linkages, plus the large number of concepts usually included.

Disadvantages of Semantic Networks

- Semantic networks have difficulty associating procedural knowledge with the facts represented by the network only that may require extensive search operations to reach conclusions.

Frames

- A frame is a knowledge representation technique that are used to represent knowledge using numbers of frames related to each other by relationship .
- Actually, a frame is a collection of attributes or slot identifying the condition of an object.
- An instance of frame is representation of a specific object.
- Every slot has name and corresponding value.
- It works based on the concept of inheritance.

Frames

- Frame is the type of schema used in many AI applications including vision and natural language processing.
- Frames are used to represent the commonsense knowledge
- Slots are similar to attributes in object oriented approach, which includes declarative as well as procedural information.

Frames

- Frame system usually have collection of frames connected to each other.
- Value of an attribute of one frame may be another frame.

Frames Example

Structure of frame:

Name of Frame	
Name of Slots	Value of Slots

Example

Car	
Engine	
Number of Doors	4

Frames Example

Three types of frames can be found in frame system:

- Class frame
- Sub class frame
- Instance frame

Class: Vehicle	
Reg. No.	
Model	
Producer	
Owner	

Class frame

Frames Example

Car	
Class: Vehicle	
Reg. No.	
Model	
Producer	
Owner	
No. of doors	4
Engine	

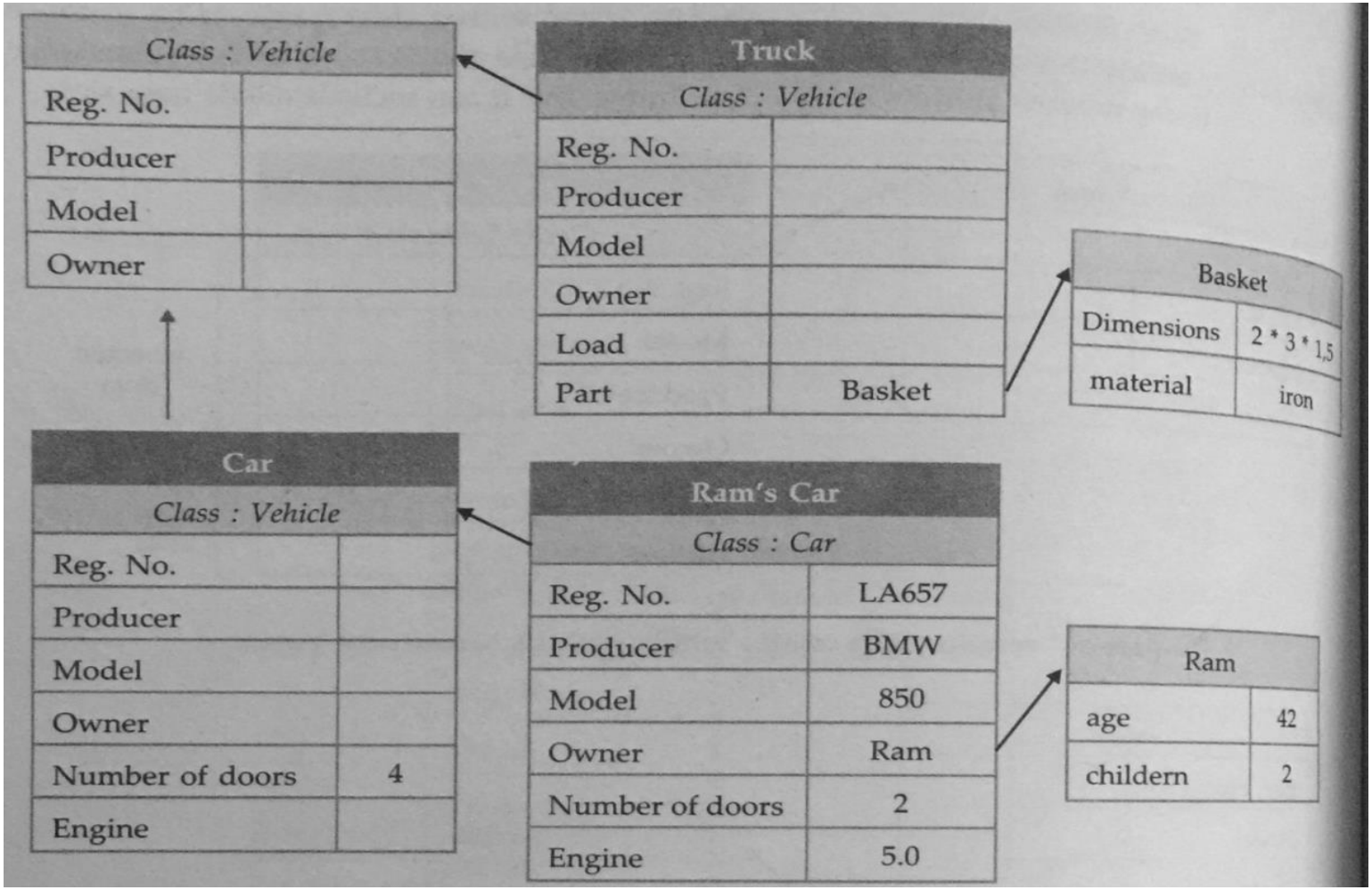
Sub-Class frame

Frames Example

Car	
Class: Vehicle	
Reg. No.	HY-24025
Model	Energy Efficient
Producer	Hyundai
Owner	Ram
No. of doors	4
Engine	Petrol

Instance frame

Frames Example



Advantages of Frame

- Makes Programming easier by grouping related knowledge.
- Frames are expressive compared to semantic nets.
- Easy to set-up slots for new properties and relations.

Disadvantages of Frame

- No standards for filling the slot values.
- More general representation rather than specific representation.
- Different frames may required to represent the same real world entity i.e. frame representation of a classroom may be different for a professor and for the maintenance staffs.
- Reasoning will be complex compared to semantic nets.

Scripts

- Scripts are a development of the idea of representing actions and events using semantic networks.
- With scripts the idea is that whole set of actions fall into stereotypical (similar tune) patterns.
- Scripts make use of the idea of defaults, with a class defining the roles in some action, and individual instances of the class having the roles filled in.
- Script has been suggested as a way of analysing complete stories.

Scripts Example

- For example, consider a story.

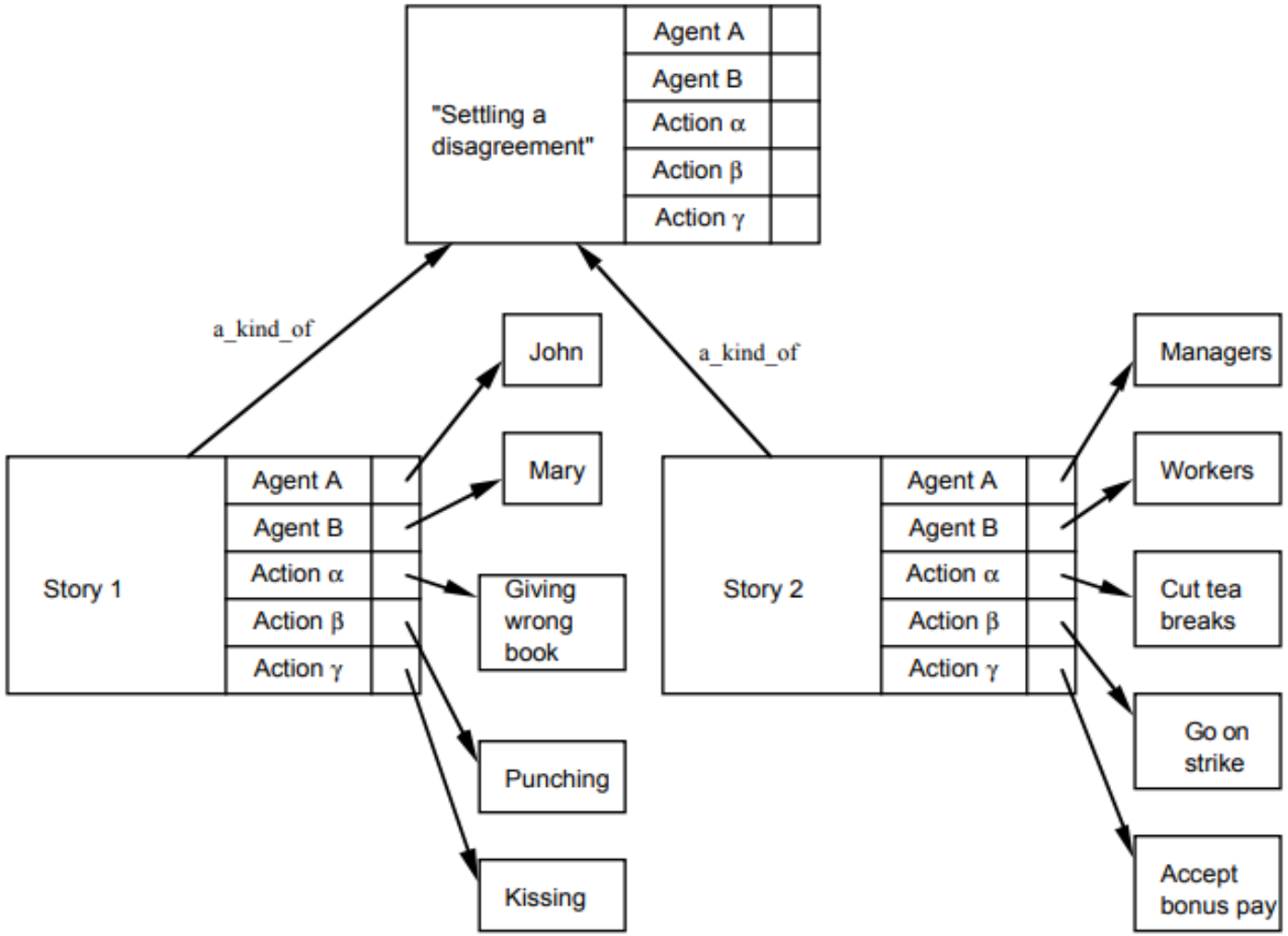
  "John gave Mary a book. The book was not the one Mary likes, so she punched John. That made her feel sorry for him, so she then kissed him".

  This may be considered an instance of the script "A did action α for B. B didn't like α, so he/she/it/they did action β to hurt A. B then came to an agreement with A and did action γ to make up".

Scripts Example

- In this example, A was John, B was Mary, α was giving the wrong book, β was punching, and γ was kissing.
- In another instance of the same script, *A could be the factory managers, B the factory workers, α could be cutting tea-break time, β could be going on strike, and γ could be agreeing to accept a bonus payment.*

Scripts Example



Conceptual Dependency

- Conceptual Dependency originally developed to represent knowledge acquired from natural language input.
- It is independent of the language.
- CD representation of a sentence is not built using words in the sentence rather built using conceptual primitives which give the intended meanings of words.
- CD provides structures and specific set of primitives from which representation can be built.

Primitives of CD

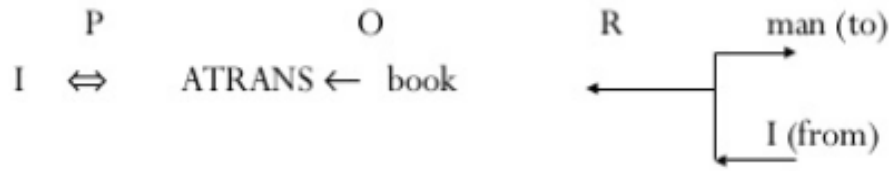
- ATRANS :Transfer of an abstract relationship (i.e. give)
- PTRANS: Transfer of the physical location of an object (e.g., go)
- PROPEL: Application of physical force to an object (e.g. push)
- MOVE :Movement of a body part by its owner (e.g. kick)
- GRASP: Grasping of an object by an action (e.g. throw)
- INGEST: Ingesting of an object by an animal (e.g. eat)
- EXPEL: Expulsion of something from the body of an animal (e.g. cry)
- MTRANS: Transfer of mental information (e.g. tell)
- MBUILD: Building new information out of old (e.g decide)
- SPEAK: Producing of sounds (e.g. say)
- ATTEND :Focusing of a sense organ toward a stimulus (e.g. listen)

Conceptual Category

- There are four conceptual categories
  - ACT Actions {one of the CD primitives}
  - PP Objects {picture producers}
  - AA Modifiers of actions {action aiders}
  - PA Modifiers of PP's {picture aiders}

Example

- I gave a book to the man



- It should be noted that this representation is same for different saying with same meaning.
- For example:
  - I gave the man a book.
  - The man got book from me.
  - The book was given to man by me etc.

Example

- Few conventions are:
  - Arrows indicate directions of dependency
  - Double arrow indicates two way link between actor and action.
  - O – for the object case relation
  - R – for the recipient case relation
  - P – for past tense
  - D – destination

Some Rules in CD

Rule 1: PP ⇔ ACT

- It describes the relationship between an actor and the event he or she causes.
  - This is a two-way dependency, since neither actor nor event can be considered primary.
  - The letter P in the dependency link indicates past tense.
- Example: John ran

CD Rep:

P

John ⇔ PTRANS

Some Rules in CD

Rule 2: ACT ← PP

- It describes the relationship between a ACT and a PP (object) of ACT.
  - The direction of the arrow is toward the ACT since the context of the specific ACT determines the meaning of the object relation.

- Example: John pushed the bike

CD Rep: John ⇔ PROPEL ← bike

Some Rules in CD

Rule 3: PP ↔ PP

- It describes the relationship between two PP's, one of which belongs to the set defined by the other.

- Example: John is doctor

CD Rep: John ↔ doctor

Some Rules in CD

Rule 4: PP ← PP

- It describes the relationship between two PP's, one of which provides a particular kind of information about the other.
  - The three most common types of information to be provided in this way are possession (shown as POSS-BY), location (shown as LOC), and physical containment (shown as CONT).
  - The direction of the arrow is again toward the concept being described.

- Example: John's dog

  CD Rep dog $\overset{\text{poss-by}}{\leftarrow}$ John

Some Rules in CD

Rule 5: PP $\Leftrightarrow$ PA

- It describes the relationship between a PP and a PA that is asserted to describe it.
  - PA represents states of PP such as height, health etc.

- Example: John is fat

CD Rep John $\Leftrightarrow$ weight (> 80)

Advantages & Disadvantages of CD

Advantages of CD:

- Using these primitives involves fewer inference rules.
- Many inference rules are already represented in CD structure.
- The holes in the initial structure help to focus on the points still to be established.

Disadvantages of CD:

- Knowledge must be decomposed into fairly low level primitives.
- Impossible or difficult to find correct set of primitives.
- A lot of inference may still be required.
- Representations can be complex even for relatively simple actions. Consider: *Dave bet Frank five pounds that Wales would win the Rugby World Cup*.
- Complex representations require a lot of storage